Strong direct product for entanglement-assisted one-way quantum communication

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Editorial note. Faithful editable TeX collection of formal author Theorem1, complete Section3 anchoring/distribution/protocol construction and Lemma30/31/32 proofs including all information, projector, unitary, channel and final coordinate-selection continuation. Required exact probability/CQ-state facts, local conditioning and substate proof, communication model and prior compression/embedding facts are retained with their original conditions. Original Figure1 is cropped from the official PDF as a diagram asset, not a source-page substitute. Full published bibliography retained. Typographic macros expand to the original register bundles. Author mathematical expressions and conditions are uncorrected; no new proof or assumptions. Only official CC BY4.0 content used; arxiv2008.08963/ECCC are identity aliases, and related arxiv2106.04299 is not an alias.

Allowed standard prerequisites (editorial).

- Quantum substate, Uhlmann and Quantum Raz lemmas. Exact author Facts13–24 and stated hypotheses retained.
- Measurement-based message compression. Exact Fact25[29] with full author explanation of the adapted register/entropy formulation retained.
- Classical embedding and minimax. Exact Facts10/12[19,25], Fact28[45] and local conditioning proofs retained.

Author Section 2.1: probability theory

The probability distribution of X is P_X, and conditioned on an event E it is $P_{X|E}$. $P_{X|Y=y}$ may be abbreviated $P_{X|y}$. The product notation $P_{XY}P_{Z|X}$ means $(P_{XY}P_{Z|X})(x,y,z) = P_{XY}(x,y)P_{Z|X=x}(z)$. The ℓ_1 distance is $\|P_X - P_{X'}\|_1 = \sum_x |P_X(x) - P_{X'}(x)|$.

Fact 4. For joint distributions on the same sets, $\|P_X - P_{X'}\|_1 \leq \|P_{XY} - P_{X'Y'}\|_1$.

Fact 5. For an event E, $|P_X(E) - P_{X'}(E)| \le \frac{1}{2}\|P_X - P_{X'}\|_1$.

Fact 6. For a coupling $P_{XX'}$ with marginals $P_X, P_{X'}$, $\|P_X - P_{X'}\|_1 \le 2P_{XX'}(X \neq X')$.

Fact 7. If $\|P_X - P_{X'}\|_1 \le \epsilon$ and $P_X(E) \ge \alpha > \epsilon$, then $\|P_{X|E} - P_{X'|E}\|_1 \le 2\epsilon/\alpha$.

Proof. Fact5 gives $\alpha - \epsilon/2 \le P_{X'}(E) \le \alpha + \epsilon/2$. There is E' with $2(P_{X|E}(E') - P_{X'|E}(E')) = \|P_{X|E} - P_{X'|E}\|_1$. Also $P_X(E \wedge E') = P_X(E)P_{X|E}(E') \ge \alpha P_{X|E}(E')$, and $P_{X'}(E \wedge E') \le (\alpha + \epsilon/2)P_{X'|E}(E') \le \alpha P_{X'|E}(E') + \frac{1}{2}\|P_X - P_{X'}\|_1$. Thus

$$\begin{aligned}\|P_X - P_{X'}\|_1 &\ge 2(P_X(E \wedge E') - P_{X'}(E \wedge E')) \\ &\ge 2\alpha(P_{X|E}(E') - P_{X'|E}(E')) - \|P_X - P_{X'}\|_1 \\ &\ge \alpha\|P_{X|E} - P_{X'|E}\|_1 - \|P_X - P_{X'}\|_1,\end{aligned}$$

giving the result. □

Fact 8 ([3],Lemma16). If $P_{XY}(x, y^*) = \alpha P_X(x)$ for every x, then

$$\|P_{XYZ} - P_{XY}P_{Z|X,y^*}\|_1 \le \frac{2}{\alpha}\|P_{XYZ} - P_{XY}P_{Z|X}\|_1.$$

Corollary 9. Suppose P_{XY} and $P_{X'Y'Z'}$ satisfy $P_X(x, y^*) = \alpha P_X(x)$ for all x. Then

$$\|P_{X'Z'|y^*} - P_{X'Z'}\|_1 \le \frac{11}{\alpha}\|P_{X'Y'Z'} - P_{XY}P_{Z'|X'}\|_1.$$

Proof. Let the norm on the RHS be ϵ. Fact7 gives $\|P_{X|y^*} - P_{X'|y^*}\|_1 \le 2\epsilon/\alpha$. Let $P_{XYZ''} = P_{XY}P_{Z'|X'Y'}$. Then

$$\begin{aligned}\|P_{X'Z'} - P_{XZ''}\|_1 &= \sum_{x,z}\left|P_{X'}(x)\sum_y P_{Y'|x}(y)P_{Z'|xy}(z) - P_X(x)\sum_y P_{Y|x}(y)P_{Z'|xy}(z)\right| \\ &\le \sum_{x,y,z}|P_{X'}(x)P_{Y'|x}(y) - P_X(x)P_{Y|x}(y)|P_{Z'|xy}(z) \\ &= \|P_{X'Y'} - P_{XY}\|_1 \le \epsilon.\end{aligned}$$

Furthermore,

$$\begin{aligned}\|P_{XYZ''} - P_{XY}P_{Z''|X}\|_1 &\le \|P_{XYZ''} - P_{X'Y'Z'}\|_1 + \|P_{X'Y'Z'} - P_{XY}P_{Z'|X'}\|_1 \\ &\quad + \|P_{XY}P_{Z'|X'} - P_{XY}P_{Z''|X}\|_1 \\ &= \|P_{XY} - P_{X'Y'}\|_1 + \|P_{X'Y'Z'} - P_{XY}P_{Z'|X'}\|_1 \\ &\quad + \sum_{x,y}P_{XY}(x,y)\|P_{Z'|x} - P_{Z''|x}\|_1 \\ &\le 2\epsilon + \sum_x P_X(x)\sum_{y,z}|P_{Y|x}(y) - P_{Y'|x}(y)|P_{Z'|xy}(z) \\ &\le 2\epsilon + \sum_{x,y}|P_X(x)P_{Y|x}(y) - P_{X'}(x)P_{Y'|x}(y)| \\ &\quad + \sum_{x,y}|P_{X'}(x) - P_X(x)|P_{Y'|x}(y) \\ &\le 2\epsilon + 2\|P_{XY} - P_{X'Y'}\|_1 \le 4\epsilon.\end{aligned}$$

Combining,

$$\begin{aligned}\|P_{X'Z'|y^*} - P_{X'Z'}\|_1 &\le \|P_{X'Z'|y^*} - P_{XZ''|y^*}\|_1 + \|P_{XZ''|y^*} - P_{XZ''}\|_1 \\ &\quad + \|P_{XZ''} - P_{X'Z'}\|_1 \\ &\le \|P_{X|y^*} - P_{X'|y^*}\|_1 + \|P_{XZ''|y^*} - P_{XZ''}\|_1 \\ &\quad + \|P_{XZ''} - P_{X'Z'}\|_1 \\ &\le \frac{2\epsilon}{\alpha} + \frac{2}{\alpha}\|P_{XYZ''} - P_{XY}P_{Z''|X}\|_1 + \epsilon \\ &\le \frac{2\epsilon}{\alpha} + \frac{8\epsilon}{\alpha} + \epsilon \le \frac{11\epsilon}{\alpha}.\end{aligned}$$

The third inequality uses Lemma8, as stated by the authors. □

Fact 10 ([19],Corollary6). If $P_{TU_1\cdots U_kV} = P_T P_{U_1|T}\cdots P_{U_k|T}P_{V|TU_1\cdots U_k}$ and E is an event, then

$$\sum_{i=1}^{k}\|P_{TU_iV|E} - P_{TV|E}P_{U_i|T}\|_1 \le \sqrt{k\left(\log|\mathcal{V}| + \log\frac{1}{\Pr[E]}\right)}.$$

Definition 11 ([19]). (X, Y) is $(1-\epsilon)$-embeddable in $(X'S, Y'T)$ if there is a random variable R independent of XY, and $f_A : \mathcal{X}\times\mathcal{R}\to\mathcal{S}$, $f_B : \mathcal{Y}\times\mathcal{R}\to\mathcal{T}$, such that $\|P_{XYf_A(X,R)f_B(X,R)} - P_{X'Y'ST}\|_1 \le \epsilon$.

Fact 12 ([19,25]). If $\|P_{X'Y'R'} - P_{XY}P_{R'|X'}\|_1 \le \epsilon$ and $\|P_{X'Y'R'} - P_{XY}P_{R'|Y'}\|_1 \le \epsilon$, then (X, Y) is $(1-5\epsilon)$-embeddable in $(X'R', Y'R')$.[1]

Author Section 2.2: quantum information

The ℓ_1 distance of states is $\|\rho-\sigma\|_1 = \operatorname{Tr}\sqrt{(\rho-\sigma)^\dagger(\rho-\sigma)} = \operatorname{Tr}|\rho-\sigma|$. Their fidelity is $\mathrm{F}(\rho,\sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$.

Fact 13 (Fuchs–van de Graaf). $2(1-\mathrm{F}(\rho,\sigma)) \le \|\rho-\sigma\|_1 \le 2\sqrt{1-\mathrm{F}(\rho,\sigma)^2}$. For pure states, the authors state

$$\||\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|\|_1 = \sqrt{1-\mathrm{F}(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|)^2} = \sqrt{1-|\langle\psi|\phi\rangle|^2}.$$

Fact 14 (Uhlmann). If states on X are purified to $|\rho\rangle, |\sigma\rangle$ on XY, then $\mathrm{F}(\rho,\sigma) = \max_U |\langle\rho|\mathbf{1}_X\otimes U|\sigma\rangle|$, over unitaries on Y.

Fact 15 (Data processing). For a quantum channel $\mathcal{E}$, $\|\mathcal{E}(\rho)-\mathcal{E}(\sigma)\|_1 \le \|\rho-\sigma\|_1$ and $\mathrm{F}(\mathcal{E}(\rho),\mathcal{E}(\sigma)) \ge \mathrm{F}(\rho,\sigma)$. Entropy is $S(\rho) = -\operatorname{Tr}(\rho\log\rho)$. Relative entropy is $S(\rho\|\sigma) = \operatorname{Tr}(\rho\log\rho) - \operatorname{Tr}(\rho\log\sigma)$. Relative min-entropy is $S_\infty(\rho\|\sigma) = \min\{\lambda : \rho \le 2^\lambda\sigma\}$. These are finite only if the support of ρ is contained in that

[1] This is equivalent to Lemma2.11 in [25], stated there in relative entropy. That proof converts to these trace distances by Pinsker's inequality, justifying the form tailored to this application.

of σ, and $0 \leq S(\rho\|\sigma) \leq S_\infty(\rho\|\sigma)$. The authors define the smooth relative min-entropy by

$$S_\infty^\epsilon(\rho\|\sigma) = \inf_{\rho':\|\rho-\rho'\|_1\leq\epsilon} S(\rho'\|\sigma).$$

It can be finite even if the original supports are not contained. Although S_∞ cannot be upper-bounded by S, its smoothed version can, by the quantum substate theorem.

Fact 16 (Quantum Substate Theorem [31,23]). For states with $\mathrm{supp}(\rho) \subseteq \mathrm{supp}(\sigma)$ and $\epsilon > 0$,

$$S_\infty^\epsilon(\rho\|\sigma) \leq \frac{4S(\rho\|\sigma)}{\epsilon^2} + \log\frac{1}{1-\epsilon^2/4}.$$

Fact 17 (Pinsker). $\|\rho-\sigma\|_1 \leq \sqrt{S(\rho\|\sigma)}$.

Fact 18. If $\sigma = \epsilon\rho + (1-\epsilon)\rho'$, then $S_\infty(\rho\|\sigma) \leq \log(1/\epsilon)$.

Fact 19. If $\mathrm{supp}(\rho) \subseteq \mathrm{supp}(\phi) \subseteq \mathrm{supp}(\sigma)$, then $S_\infty(\rho\|\sigma) \leq S_\infty(\rho\|\phi) + S_\infty(\phi\|\sigma)$.

Fact 20. For unitary U, $S_\infty(U\rho U^\dagger\|U\sigma U^\dagger) = S_\infty(\rho\|\sigma)$. A CQ state is $\rho_{XY} = \sum_x P_X(x)|x\rangle\langle x|_X \otimes \rho_{Y|x}$ with classical register X and quantum Y. X also denotes the associated random variable. $\rho_{Y|x}$ is the state conditioned on observing x on X, so $\rho_{XY|x} = |x\rangle\langle x|_X \otimes \rho_{Y|x}$. If registers are clear, write ρ_x. Mutual information is $I(Y:Z)_\rho = S(\rho_{YZ}\|\rho_Y\otimes\rho_Z)$. Conditioned on classical X, $I(Y:Z|X) = \mathbb{E}_{P_X}[I(Y:Z)_{\rho_x}]$. The chain rule is $I(XY:Z)_\rho = I(X:Z)_\rho + I(Y:Z|X)_\rho$.

Fact 21 ([6],LemmaB.7). $\inf_{\sigma_Z} S_\infty(\rho_{YZ}\|\rho_Y\otimes\sigma_Z) \leq 2\min\{\log|\mathcal{Y}|, \log|\mathcal{Z}|\}$.

Fact 22. For CQ states $\rho_{XY} = \sum_x P_X(x)|x\rangle\langle x|_X\otimes\rho_{Y|x}$ and $\sigma_{XY} = \sum_x P_{X'}(x)|x\rangle\langle x|_X\otimes\sigma_{Y|x}$,

$$S(\rho_{XY}\|\sigma_{XY}) = S(P_X\|P_{X'}) + \mathbb{E}_{P_X}[S(\rho_{Y|x}\|\sigma_{Y|x})].$$

Fact 23. Let $\sigma_{XYZ} = \sum_{x,y} P_{XY}(x,y)|x,y\rangle\langle x,y|\otimes\sigma_{Z|xy}$ and $\rho_{XYZ} = \sum_{x,y} P_{X'Y'}(x,y)|x,y\rangle\langle x,y|\otimes\sigma_{Z|xy}$, with $\|P_{XY}-P_{X'Y'}\|_1 \leq \delta$ and $I(Y:Z|X)_\sigma \leq c$. For $0<\epsilon<1/4$,

$$P_{X'Y'}\left(S_\infty^\epsilon(\sigma_{Z|xy}\|\sigma_{Z|x}) > \frac{4c+1}{\epsilon^3}\right) \leq \epsilon + \delta/2.$$

Proof. $\mathbb{E}_{P_{XY}}[S(\sigma_{Z|xy}\|\sigma_{Z|x})] = I(Y:Z|X)_\sigma \leq c$. Markov gives $P_{XY}(S(\sigma_{Z|xy}\|\sigma_{Z|x}) > c/\epsilon) \leq \epsilon$. The substate theorem implies

$$\begin{aligned}&P_{XY}\left(S_\infty^\epsilon(\sigma_{Z|xy}\|\sigma_{Z|x}) > \frac{4c+1}{\epsilon^3}\right)\\ &\leq P_{XY}\left(S_\infty^\epsilon(\sigma_{Z|xy}\|\sigma_{Z|x}) > \frac{4c}{\epsilon^3} + \log\frac{1}{1-\epsilon^2/4}\right) \leq \epsilon.\end{aligned}$$

The distance assumption transfers this to $P_{X'Y'}$ with additive $\delta/2$. □

Fact 24 (Quantum Raz's Lemma [3]). For CQ ρ_{XY}, σ_{XY}, with classical $X = X_1 \cdots X_k$ and σ product across all registers, $\sum_{i=1}^{k} I(X_i : Y)_\rho \leq S(\rho_{XY} \| \sigma_{XY})$.

Fact 25 ([29],Lemma2). Suppose

$$|\sigma\rangle_{X\widetilde{X}AB} = \sum_x \sqrt{P_X(x)} |xx\rangle_{X\widetilde{X}} |\sigma\rangle_{AB|x}$$

satisfies $P_X(S_\infty^\epsilon(\sigma_{B|x} \| \sigma_B) > c) \leq \delta$ for $\delta > 0$. There are measurement operators $\{\Pi_x\}_x$ acting only on $X\widetilde{X}A$ such that: (i) each succeeds with probability $\alpha = 2^{-c/\delta}$ on this state, namely $\|(\Pi_x \otimes \mathbf{1}_B)|\sigma\rangle\|_2^2 = 2^{-c/\delta}$; (ii) $(\Pi_x \otimes \mathbf{1}_B)|\sigma\rangle\langle\sigma|(\Pi_x \otimes \mathbf{1}_B)$ is $|xx\rangle\langle xx| \otimes \rho_x$ for a state on AB, and

$$\mathbb{E}_{P_X} \left[\left\| \frac{1}{\alpha} (\Pi_x \otimes \mathbf{1}_B) |\sigma\rangle\langle\sigma|_{X\widetilde{X}AB} (\Pi_x \otimes \mathbf{1}_B) - |xx\rangle\langle xx|_{X\widetilde{X}} \otimes |\sigma\rangle\langle\sigma|_{AB|x} \right\|_1 \right] \leq \epsilon + 2\delta.$$

The authors' version differs slightly from [29]: the original uses $I(X : B)_\sigma$ and no $\widetilde{X}$. In that proof the mutual information is converted to $P_X(S_\infty^\epsilon(\sigma_{B|x} \| \sigma_B) > c)$ anyway, so the first change makes no difference. The same projector taking the state without $\widetilde{X}$ to $|x\rangle\langle x| \otimes |\sigma\rangle\langle\sigma|_{AB|x}$ also takes the state with $\widetilde{X}$ to $|xx\rangle\langle xx| \otimes |\sigma\rangle\langle\sigma|_{AB|x}$, so the second makes no difference.

Author Section 2.3: quantum communication and games

A quantum protocol for $f \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$ has Alice and Bob with inputs x, y and prior shared entanglement. Each round one party applies a unitary to its classical input, last received message and retained memory, keeps some registers and sends the rest. Safe copies of inputs can be made by CNOT so input registers emerge unchanged. Shared classical randomness is included in this model by both parties measuring shared entanglement in the same basis. In one-way communication Alice sends the single message; retained registers are discarded. Bob performs a projective measurement on his input, shared entanglement and message, yielding the output $P(x, y)$. Equivalently he performs a unitary then measures designated $\log |\mathcal{Z}|$ output qubits in the computational basis. Arbitrary entanglement and local-operation complexity are allowed; the resource is qubits sent.

Definition 26. For $0 < \epsilon < 1$, $Q_\epsilon^1(f)$ is the minimum message size of a one-way entanglement-assisted protocol with $\Pr[P(x, y) \in f(x, y)] \geq 1 - \epsilon$ for every (x, y), over inherent protocol randomness.

Definition 27. $Q_{p,\epsilon}^1(f)$ is the corresponding minimum message size when success $\geq 1 - \epsilon$ is averaged over $(x, y) \sim p$ and protocol randomness.

Fact 28 (Yao's lemma [45]). $Q^1_\epsilon(f) = \max_p Q^1_{p,\epsilon}(f)$ for $0 < \epsilon < 1$. A two-player non-local game is $G = (q, \mathcal{X} \times \mathcal{Y}, \mathcal{A} \times \mathcal{B}, V)$: the referee samples $(x, y) \sim q$, gives inputs, and accepts outputs iff $V(x, y, a, b) = 1$. Entangled players measure their respective shared-state halves and inputs (unitary followed by computational-basis measurement on designated $\log|\mathcal{A}|$, $\log|\mathcal{B}|$ output qubits).

Definition 29. The entangled value $\omega^*(G)$ is the maximum winning probability over all shared-entanglement strategies, averaged over q and inherent randomness. A distribution is anchored on one side with probability ζ if either some x^* has $q(x^*) = \zeta$, $q(y|x^*) = q(y)$ for all y, or some y^* has $q(y^*) = \zeta$, $q(x|y^*) = q(x)$ for all x (author Section1.1).

Formal author direct-product theorem

Theorem 1. For any relation $f \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$, and any $0 < \epsilon, \zeta < 1/2$,

$$Q^1_{1-(1-\epsilon)^{\Omega(\zeta^6 k/\log|\mathcal{Z}|)}}(f^k) = \Omega\big(k\left(\zeta^5 Q^1_{\epsilon+\zeta}(f) - \log\log(1/\zeta)\right)\big).$$

Author Section 3.1: setup

Let p be the hard distribution for $Q^1_{\epsilon+12\zeta}(f)$ from Yao's lemma, so $Q^1_{\epsilon+12\zeta}(f) = Q^1_{p,\epsilon+12\zeta}(f)$. Extend f to $\widetilde{f} \subseteq \mathcal{X} \times (\mathcal{Y} \cup \{y^*\}) \times \mathcal{Z}$ by including (x, y^*, z) for every x, z. Extend p with $p(y^*) = 0$. Then

$$Q^1_{p,\gamma}(\widetilde{f}) = Q^1_{p,\gamma}(f) \tag{1}$$

for every error γ. Also

$$Q^1_\gamma(f^k) \geq Q^1_\gamma(\widetilde{f}^k). \tag{2}$$

Every protocol for f^k gives one for $\widetilde{f}^k$: on positions with y^* Bob pretends an input from $\mathcal{Y}$, runs it and answers accordingly. Its output is correct there regardless. We show, for the distribution q below,

$$Q^1_{q^k,1-(1-\epsilon)^{\Omega(\zeta^6 k/\log|\mathcal{Z}|)}}(\widetilde{f}^k) \geq \frac{\zeta^5 k}{60} Q^1_{p,\epsilon+12\zeta}(\widetilde{f}) - k\log\log\frac{24}{5\zeta}. \tag{3}$$

Since $Q^1_\gamma(\widetilde{f}^k) \geq Q^1_{q^k,\gamma}(\widetilde{f}^k)$, (1)–(3) imply Theorem1. Set $q(x, y) = (1-\zeta)p(x, y)$ for $y \in \mathcal{Y}$ and $q(x, y^*) = \zeta p(x)$. Clearly $q(x, y^*) = q(x)q(y^*)$ and

$$\|p(x, y) - q(x, y)\|_1 \leq 2\zeta. \tag{4}$$

For each $i \in [k]$, define independent copies of $X_i Y_i D_i G_i$, with marginal $X_i Y_i$ equal to q. $D_i G_i$ are correlation-breaking variables: given their value, X_i, Y_i are independent. D_i is uniform in $\{0, 1\}$. Its conditional G_i distribution is:

- If $D_i = 0$, $G_i = x$ w.p. $p(x)$.
- If $D_i = 1$, $G_i = y^*$ w.p. $1 - (1-\zeta)^{2/3}$, and $G_i = y$ w.p. $(1-\zeta)^{2/3} p(y)$.

Conditional $X_i Y_i$ distributions are:

- If $D_i = 0, G_i = x$: (x, y^*) has probability ζ, and (x, y) has probability $(1-\zeta)p(y|x)$.
- If $D_i = 1, G_i = y^*$: (x, y^*) has probability $p(x)$.
- If $D_i = 1, G_i = y$: (x, y^*) has probability $(1-(1-\zeta)^{1/3})p(x|y)$, and (x, y) has probability $(1-\zeta)^{1/3}p(x|y)$.

Lemma 30. $P_{X_iY_i}(x, y) = q(x, y)$ for every $(x, y) \in \mathcal{X} \times (\mathcal{Y} \cup \{y^*\})$.

Proof. For $D_i = 0$, $G_i = X_i$, so $P_{G_iY_i|D_i=0} = P_{X_iY_i|D_i=0} = q$. For $D_i = 1$ and $y \in \mathcal{Y}$, $Y_i = y$ only if $G_i = y$, hence

$$\begin{aligned} P_{X_iY_i|D_i=1}(x, y) &= P_{G_i|D_i=1}(y)P_{X_iY_i|D_i=1,G_i=y}(x, y) \\ &= (1-\zeta)^{2/3}p(y)(1-\zeta)^{1/3}p(x|y) = (1-\zeta)p(x, y) = q(x, y). \end{aligned}$$

$Y_i = y^*$ can arise when $G_i = y^*$ or any $y \in \mathcal{Y}$, so

$$\begin{aligned} P_{X_iY_i|D_i=1}(x, y^*) &= P_{G_i|D_i=1}(y^*)P_{X_iY_i|D_i=1,G_i=y^*}(x, y^*) \\ &\quad + \sum_y P_{G_i|D_i=1}(y)P_{X_iY_i|D_i=1,G_i=y}(x, y^*) \\ &= (1-(1-\zeta)^{2/3})p(x) \\ &\quad + (1-\zeta)^{2/3}(1-(1-\zeta)^{1/3})\sum_y p(y)p(x|y) \\ &= (1-(1-\zeta)^{2/3})p(x) + ((1-\zeta)^{2/3} - (1-\zeta))p(x) \\ &= \zeta p(x) = q(x, y^*). \end{aligned}$$

□

In particular $P_{X_iY_i}(x, y^*) = P_{X_i}(x)P_{Y_i}(y^*)$. Also

$$P_{Y_iG_i|D_i=1}(Y_i \neq G_i) = (1-\zeta)^{2/3}(1-(1-\zeta)^{1/3}) \leq 1 - 2\zeta/3 - 1 + \zeta = \zeta/3. \quad (5)$$

Let $\mathcal{P}$ be a one-way protocol for $\widetilde{f}^k$ of cost ck. Alice/Bob's inputs are $X = X_1 \cdots X_k$, $Y = Y_1 \cdots Y_k$. They share an input-uncorrelated entangled pure state on $E^A E^B$. Alice applies V^A on XE^A, producing message M and discarded register A. Let $|\theta\rangle_{AME^B|x}$ denote that pure state; it only depends on y via x. For input distribution P_{XY} the CQ state is

$$\theta_{XYAME^B} = \sum_{xy} P_{XY}(xy)|xy\rangle\langle xy|_{XY} \otimes |\theta\rangle\langle\theta|_{AME^B|x}.$$

Purify it using $\widetilde{X}, \widetilde{Y}$:

$$|\theta\rangle_{X\widetilde{X}Y\widetilde{Y}AME^B} = \sum_{xy} \sqrt{P_{XY}(xy)}|xxyy\rangle_{X\widetilde{X}Y\widetilde{Y}}|\theta\rangle_{AME^B|x}.$$

Bob applies V^B on YME^B, converting ME^B to BZ, with output $Z = Z_1 \cdots Z_k$. Call the resulting pure state $|\rho\rangle_{X\widetilde{X}Y\widetilde{Y}ABZ}$. Define P_{XYDGZ} with the earlier $XYDG$, Z independent of DG given XY, and its conditional law from measuring Z in $|\rho\rangle$.

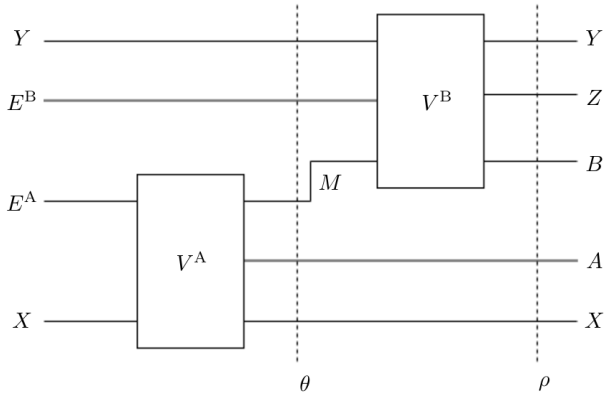


Figure1. One-way quantum protocol $\mathcal{P}$.

Author Section 3.2: proof of Theorem 1

We show that if $ck < \zeta^5 k Q^1_{p,\epsilon+12\zeta}(\widetilde{f})/300 - k\log\log(24/(5\zeta))$, the success of $\mathcal{P}$ is $(1-\epsilon)^{\Omega(\zeta^6 k/\log|\mathcal{Z}|)}$.

Lemma 31. Let $\delta = \zeta^6/1440000$, $\delta' = \zeta^6/(1440000\log|\mathcal{Z}|)$. Let T_i indicate correctness of $\mathcal{P}$ on $f(X_i, Y_i)$. If its communication is $< \zeta^5 k Q^1_{p,\epsilon+12\zeta}(\widetilde{f})/60 - k\log\log(24/(5\zeta))$, there exist $\lfloor\delta' k\rfloor$ coordinates $i_1, \ldots, i_{\lfloor\delta' k\rfloor}$ such that for all $1 \le r \le \lfloor\delta' k\rfloor - 1$ either

$$\Pr[\prod_{j=1}^{r} T_{i_j} = 1] \le (1-\epsilon)^{\delta k},$$

or $\Pr[T_{i_{r+1}} = 1 \mid \prod_{j=1}^{r} T_{i_j} = 1] \le 1-\epsilon$. Inductively suppose $C = \{i_1, \ldots, i_t\}$ has been found, $1 \le t \le \lfloor\delta' k\rfloor$, with each conditional success at most $1-\epsilon$. Let $\mathcal{E}$ be $\prod_{i\in C} T_i = 1$. If $\Pr[\mathcal{E}] \le (1-\epsilon)^{\delta k}$, done. Otherwise identify the next coordinate with conditional success at most $1-\epsilon$. The first coordinate is similar with empty conditioning; only the lower bound on conditioning probability is used. Use $|\phi\rangle$, the state $|\rho\rangle$ conditioned on $\mathcal{E}$. For each $DG = dg$ define

$$|\phi\rangle_{X\widetilde{X}Y\widetilde{Y}ABZ|dg} = \frac{1}{\sqrt{\gamma_{dg}}} \sum_{xy} \sqrt{P_{XY|dg}(xy)}|xxyy\rangle_{X\widetilde{X}Y\widetilde{Y}} \otimes \sum_{z_C:(x_C,y_C,z_C)\in\widetilde{f}^t} |z_C\rangle_{Z_C}|\widetilde{\phi}\rangle_{ABZ_{\overline{C}}|xyz_C}.$$

The subnormalized state obeys $\||\widetilde{\phi}\rangle_{ABZ_{\overline{C}}|xyz_C}\|_2^2 = P_{Z_C|xy}(z_C)$. $\gamma_{dg} = \Pr[\mathcal{E}|dg]$ and $\sum_{dg} P_{DG}(dg)\gamma_{dg} = \Pr[\mathcal{E}]$. The distribution of XYZ in this state is $P_{XYZ|\mathcal{E},dg}$. Registers DG are not explicitly considered: $|\phi\rangle_{dg}$ is notation without measuring such registers. Likewise $|\phi\rangle_{d_{-i}g_{-i}}$ conditions on $d_{-i}g_{-i}$ and changes only normalization. $\phi_{x_i y_i d_{-i} g_{-i}}$ is obtained by measurement of the actual $X_i Y_i$ registers. For $i \notin \overline{C}$, the authors use pure states $|\phi\rangle_{X_{\overline{C}}\widetilde{X}_{\overline{C}}Y_{\overline{C}}\widetilde{Y}_{\overline{C}}ABZ_{\overline{C}}|x_i y_i x_C y_C z_C d_{-i} g_{-i}}$.

Lemma 32. If $\Pr[\mathcal{E}] \geq (1-\epsilon)^{\delta k}$, there is $i \in \overline{C}$, $R_i = X_C Y_C Z_C D_{-i} G_{-i}$, and for each $R_i = r_i$ a state $|\phi'\rangle_{X_{\overline{C}}\widetilde{X}_{\overline{C}}Y_{\overline{C}}\widetilde{Y}_{\overline{C}}ABZ_{\overline{C}}|y^* r_i}$ satisfying:

(i) $\|P_{X_i Y_i R_i|\mathcal{E}} - P_{X_i Y_i} P_{R_i|\mathcal{E},X_i}\|_1 \leq 7\zeta/120$.

(ii) $\|P_{X_i Y_i R_i|\mathcal{E}} - P_{X_i Y_i} P_{R_i|\mathcal{E},Y_i}\|_1 \leq 7\zeta/120$.

(iii) There are projectors $\Pi_{x_i r_i}$ on $X_{\overline{C}}\widetilde{X}_{\overline{C}}A$ and unitaries $U_{y_i r_i}$ on $Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}}$. Each projector succeeds on $|\phi'\rangle_{r_i}$ with probability $\alpha = 2^{-c'}$, $c' \leq 60c/\zeta^5$, and

$$\mathbb{E}_{P_{X_i Y_i R_i|\mathcal{E}}} \Big[\Big\| \frac{1}{\alpha} (\Pi_{x_i r_i} \otimes U_{y_i r_i}) |\phi'\rangle\langle\phi'|_{y^* r_i} (\Pi_{x_i r_i} \otimes U_{y_i r_i}^\dagger) - |\phi\rangle\langle\phi|_{x_i y_i r_i} \Big\|_1 \Big] \leq 21\zeta.$$

Its complete author proof is in Section3.3 below.

Proof of Lemma 31. Construct a protocol $\mathcal{P}'$ for $\widetilde{f}$ on distribution $P_{X_i Y_i} = q$ by embedding inputs into the chosen state.

- Alice/Bob have r distributed as required by Fact12 as shared randomness, and $2^{60c/\zeta^5}\log(24/(5\zeta))$ copies of $|\phi'\rangle_{y^* r_i}$ as entanglement, Alice holding $X_{\overline{C}}\widetilde{X}_{\overline{C}}A$ and Bob the remaining registers.
- On input (x_i, y_i), Fact12 and Lemma32(i),(ii) give R_i^A, R_i^B with
$$\|P_{X_i Y_i R_i^A R_i^B} - P_{X_i Y_i R_i R_i|\mathcal{E}}\|_1 \leq 7\zeta/24,$$
where $R_i R_i$ are perfectly correlated copies.
- Alice measures $\{\Pi_{x_i r_i^A}, 1 - \Pi_{x_i r_i^A}\}$ on each copy. If none succeeds, abort. Otherwise send a $(60c/\zeta^5 + \log\log(24/(5\zeta)))$-bit message indexing a successful copy.
- Bob applies $U_{y_i r_i^B}$ on that copy and measures Z_i for his output.

By construction,

$$\mathbb{E}_{P_{X_i Y_i R_i|\mathcal{E}}} \Pr[Z_i \text{ measured on } |\phi\rangle_{x_i y_i r_i} \in \widetilde{f}(x_i, y_i)] = \Pr[T_i = 1|\mathcal{E}].$$

First suppose the inputs and R_i^A, R_i^B follow exactly $P_{X_i Y_i R_i R_i|\mathcal{E}}$, writing both copies r_i. The abort probability is

$$(1 - 2^{-c'})^{2^{60c/\zeta^5}\log(24/(5\zeta))} \leq 5\zeta/24.$$

If not aborting, the state after Bob's unitary is $\alpha^{-1/2}(\Pi_{x_i r_i} \otimes U_{y_i r_i})|\phi'\rangle_{y^* r_i}$. Lemma32(iii) gives expected correctness at least $\Pr[T_i = 1|\mathcal{E}] - 21\zeta/2$. Including abort, at least $\Pr[T_i = 1|\mathcal{E}] - 21\zeta/2 - 5\zeta/24$. For the actual R_i^A, R_i^B distribution, success is at least

$$\Pr[T_i = 1|\mathcal{E}] - 21\zeta/2 - 5\zeta/24 - 7\zeta/24 \geq \Pr[T_i = 1|\mathcal{E}] - 11\zeta.$$

Since $\|q - p\|_1 \leq 2\zeta$, running this protocol on p gives success $\geq \Pr[T_i = 1|\mathcal{E}] - 12\zeta$. Communication is $60c/\zeta^5 + \log\log(24/(5\zeta)) < Q^1_{p,\epsilon+12\zeta}(\widetilde{f})$. If $\Pr[T_i = 1|\mathcal{E}] \geq 1 - \epsilon$, error is $\leq \epsilon + 12\zeta$, a contradiction. Hence $\Pr[T_i = 1|\mathcal{E}] \leq 1 - \epsilon$. Take $i_{t+1} = i$. □

Complete author Section 3.3, distribution bounds

First show in expectation over $i \in \overline{C}$ that distributions conditioned on $\mathcal{E}$ are close to unconditional ones. Fact10 with trivial T, V and $U_i = X_i Y_i D_i G_i$ gives

$$\mathbb{E}_{i\in\overline{C}} \| P_{X_i Y_i D_i G_i|\mathcal{E}} - P_{X_i Y_i D_i G_i} \|_1 \le \frac{1}{k-t}\sqrt{k \log((1-\epsilon)^{-\delta k})} \le \sqrt{2\delta}. \tag{6}$$

By(5),

$$\mathbb{E}_{i\in\overline{C}} P_{Y_i G_i|\mathcal{E},D_i=1}(Y_i = G_i) \ge 1 - \zeta/3 - \sqrt{2\delta}. \tag{7}$$

Since $P_{G_i|D_i=1}(y^*) = 1-(1-\zeta)^{2/3}$, and $P_{Y_i|D_i=1,G_i=y_i}(y_i) = (1-\zeta)^{1/3}$ for $y_i \in \mathcal{Y}$,

$$\begin{aligned} \zeta + \sqrt{2\delta} &\ge 1 - (1-\zeta)^{2/3} + \sqrt{2\delta} \\ &\ge \mathbb{E}_{i\in\overline{C}} P_{G_i|\mathcal{E},D_i=1}(y^*) \\ &\ge 1 - (1-\zeta)^{2/3} - \sqrt{2\delta} \ge 2\zeta/3 - \sqrt{2\delta}, \end{aligned} \tag{8}$$

and

$$\begin{aligned} (1 - \zeta/3 + \sqrt{2\delta})\mathbb{E}_i P_{G_i|\mathcal{E},D_i=1}(y_i) &\ge \mathbb{E}_i P_{Y_i G_i|\mathcal{E},D_i=1}(y_i, y_i) \\ &\ge (1 - \zeta - \sqrt{2\delta})\mathbb{E}_i P_{G_i|\mathcal{E},D_i=1}(y_i). \end{aligned} \tag{9}$$

Use Fact10 again with $U_i = X_i Y_i$, $T = X_C Y_C DG$, $V = Z_C$. Set $\delta_1 = \delta + \delta' \log|\mathcal{Z}| = \zeta^6/720000$. Then

$$\begin{aligned} \sqrt{2\delta_1} &\ge \mathbb{E}_i \| P_{X_i Y_i X_C Y_C Z_C DG|\mathcal{E}} - P_{X_C Y_C Z_C DG|\mathcal{E}} P_{X_i Y_i|X_C Y_C DG} \|_1 \\ &= \mathbb{E}_i \| P_{X_i Y_i X_C Y_C Z_C DG|\mathcal{E}} - P_{X_C Y_C Z_C DG|\mathcal{E}} P_{X_i Y_i|D_i G_i} \|_1 \\ &= \mathbb{E}_i \| P_{X_i Y_i D_i G_i R_i|\mathcal{E}} - P_{D_i G_i R_i|\mathcal{E}} P_{X_i Y_i|D_i G_i} \|_1. \end{aligned} \tag{10}$$

D_i is still uniform even conditioned on $\mathcal{E}$. From(10),

$$\begin{aligned} \sqrt{2\delta_1} &\ge \tfrac{1}{2}\mathbb{E}_i \| P_{X_i Y_i G_i R_i|\mathcal{E},D_i=0} - P_{G_i R_i|\mathcal{E},D_i=0} P_{X_i Y_i|G_i,D_i=0} \|_1 \\ &= \tfrac{1}{2}\mathbb{E}_i \| P_{X_i Y_i R_i|\mathcal{E}} - P_{X_i R_i|\mathcal{E}} P_{Y_i|X_i} \|_1, \end{aligned}$$

since $X_i = G_i$ if $D_i = 0$. Combine with $\mathbb{E}_i \| P_{X_i|\mathcal{E}} - P_{X_i} \|_1 \le \sqrt{2\delta}$:

$$\mathbb{E}_i \| P_{X_i Y_i R_i|\mathcal{E}} - P_{X_i Y_i} P_{R_i|\mathcal{E},X_i} \|_1 \le 3\sqrt{2\delta_1} < 7\zeta^3/600. \tag{11}$$

Corollary9 gives

$$\mathbb{E}_i \| P_{X_i R_i|\mathcal{E},y^*} - P_{X_i R_i|\mathcal{E}} \|_1 \le 33\sqrt{2\delta_1}/\zeta. \tag{12}$$

Let F_i be $Y_i = G_i$. From(7), its conditional probability is at least $1 - \zeta/3 - \sqrt{2\delta}$ in expectation. Fact7 gives

$$\begin{aligned} &\mathbb{E}_i \| P_{X_i Y_i R_i|\mathcal{E}} - P_{Y_i R_i|\mathcal{E}} P_{X_i|Y_i} \|_1 \\ &= \mathbb{E}_i \| P_{X_i Y_i G_i R_i|\mathcal{E},D_i=1,F_i} - P_{G_i R_i|\mathcal{E},D_i=1,F_i} P_{X_i Y_i|G_i D_i=1,F_i} \|_1 \\ &\le 6\mathbb{E}_i \| P_{X_i Y_i D_i R_i|\mathcal{E},D_i=1} - P_{G_i R_i|\mathcal{E},D_i=1} P_{X_i Y_i|G_i D_i=1} \|_1 \le 6\sqrt{2\delta_1}. \end{aligned}$$

Using $\mathbb{E}_i \| P_{Y_i|\mathcal{E}} - P_{Y_i} \|_1 \le \sqrt{2\delta}$ as before,

$$\mathbb{E}_i \| P_{X_i Y_i R_i|\mathcal{E}} - P_{X_i Y_i} P_{R_i|\mathcal{E},Y_i} \|_1 \le 7\sqrt{2\delta_1} = 7\zeta^3/600. \tag{13}$$

Complete author Lemma32 proof, projectors

We show projectors $\Pi_{x_i r_i}$ take a soon-defined $|\phi'\rangle_{y^* r_i}$ close to $|\phi\rangle_{x_i y^* r_i}$. The message M has ck qubits. By Fact21 for each $DG = dg$ there is $\sigma_{M|dg}$ with

$$S_\infty(\theta_{XY\widetilde{Y}E^B M|dg} \| \theta_{XY\widetilde{Y}E^B|dg} \otimes \sigma_{M|dg}) \leq 2ck.$$

By Fact20,

$$S_\infty(\rho_{XY\widetilde{Y}BZ|dg} \| V^B(\theta_{XY\widetilde{Y}E^B|dg} \otimes \sigma_{M|dg})(V^B)^\dagger) \leq 2ck.$$

Let

$$\psi_{X_{\overline{C}}Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}}|dg} := \mathrm{Tr}_{Z_C}\left(V^B(\theta_{XYE^B|dg} \otimes \sigma_{M|x_C y_C dg})(V^B)^\dagger\right).$$

$\theta_{XY\widetilde{Y}E^B|dg} \otimes \sigma_{M|dg}$ is product across X and the rest; V^B does not act on X. Hence ψ is product across X and the remaining registers, and all X_i are mutually product. We have

$$S_\infty(\rho_{XY\widetilde{Y}BZ_{\overline{C}}|dg} \| \psi_{XY\widetilde{Y}BZ_{\overline{C}}|dg}) \leq 2ck.$$

Facts22/19 imply

$$\begin{aligned}
&\mathbb{E}_{P_{X_C Y_C Z_C DG|\mathcal{E}}} S(\phi_{X_{\overline{C}}Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}}|x_C y_C z_C dg} \| \psi_{X_{\overline{C}}Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}}|x_C y_C dg}) \\
&\leq \mathbb{E}_{P_{Z_C DG|\mathcal{E}}} S(\phi_{XY\widetilde{Y}BZ_{\overline{C}}|z_C dg} \| \psi_{XY\widetilde{Y}BZ_{\overline{C}}|dg}) \\
&\leq \mathbb{E}_{P_{Z_C DG|\mathcal{E}}} S_\infty(\phi_{XY\widetilde{Y}BZ_{\overline{C}}|z_C dg} \| \psi_{XY\widetilde{Y}BZ_{\overline{C}}|dg}) \\
&\leq \mathbb{E}_{P_{Z_C DG|\mathcal{E}}} \Big[S_\infty(\phi_{XY\widetilde{Y}BZ_{\overline{C}}|z_C dg} \| \phi_{XY\widetilde{Y}BZ_{\overline{C}}|dg}) \\
&\qquad + S_\infty(\phi_{XY\widetilde{Y}BZ_{\overline{C}}|dg} \| \rho_{XY\widetilde{Y}BZ_{\overline{C}}|dg}) \\
&\qquad + S_\infty(\rho_{XY\widetilde{Y}BZ_{\overline{C}}|dg} \| \psi_{XY\widetilde{Y}BZ_{\overline{C}}|dg}) \Big] \\
&\leq \mathbb{E}_{P_{Z_C DG|\mathcal{E}}} [\log(1/P_{Z_C|\mathcal{E}}(z_C)) + \log(1/\Pr[\mathcal{E}]) + 2ck] \\
&\leq |C| \log |\mathcal{Z}| + \delta k + 2ck \leq (\delta_1 + 2c)k.
\end{aligned}$$

By Quantum Raz's Lemma,

$$\begin{aligned}
4c + 2\delta_1 &\geq \mathbb{E}_i \mathbb{E}_{P_{X_C Y_C Z_C DG|\mathcal{E}}} I(X_i : Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}})_{\phi_{x_C y_C z_C dg}} \\
&= \mathbb{E}_i \mathbb{E}_{P_{D_i G_i R_i|\mathcal{E}}} I(X_i : Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}})_{\phi_{d_i g_i r_i}} \\
&\geq \mathbb{E}_i \tfrac{1}{2} P_{G_i|\mathcal{E},D_i=1}(y^*) \mathbb{E}_{P_{R_i|\mathcal{E},D_i=1,G_i=y^*}} I(X_i : Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}})_{\phi_{r_i|D_i=1,G_i=y^*}} \\
&\geq \mathbb{E}_i \tfrac{1}{2}(2\zeta/3 - \sqrt{2\delta}) \mathbb{E}_{P_{R_i|\mathcal{E},D_i=1,G_i=y^*}} I(X_i : Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}})_{\phi_{r_i,D_i=1,G_i=y^*}}.
\end{aligned} \tag{14}$$

Use(8) in the last step. The state $\phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}Y_{\overline{C}}\widetilde{Y}_{\overline{C}}ABZ_{\overline{C}}|x_i r_i, D_i=1, G_i=y^*}$ is the same as $\phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}Y_{\overline{C}}\widetilde{Y}_{\overline{C}}ABZ_{\overline{C}}|x_i y^* r_i}$, where Y_i, rather than G_i, is conditioned. $|\phi\rangle_{r_i, D_i=1, G_i=y^*}$ is the superposition over X_i of those states with distribution $P_{X_i|\mathcal{E},r_i,D_i=1,G_i=y^*}$. It differs from $|\phi\rangle_{y^* r_i}$ only in that distribution, the latter using $P_{X_i|\mathcal{E},y^* r_i}$. Abbreviate the former as $|\phi\rangle_{r_i,1,y^*}$; this is distinct from $|\phi\rangle_{y^* r_i}$. Use the same

convention for distributions. $P_{X_i|1,y^*} = P_{X_i|y^*}$ and $P_{R_i|\mathcal{E},x_i,1,y^*} = P_{R_i|\mathcal{E},x_iy^*}$. Hence

$$\begin{aligned}
&\mathbb{E}_i\|P_{X_iR_i|\mathcal{E},y^*} - P_{X_iR_i|\mathcal{E},1,y^*}\|_1 \\
&\le \mathbb{E}_i[\|P_{X_iR_i|\mathcal{E},y^*} - P_{X_i|y^*}P_{R_i|\mathcal{E},X_i,y^*}\|_1 \\
&\qquad + \|(P_{X_i|1,y^*} - P_{X_i|\mathcal{E},1,y^*})P_{R_i|\mathcal{E},X_i,y^*}\|_1] \\
&\le \mathbb{E}_i\left[\frac{\|P_{X_iR_i|\mathcal{E}} - P_{X_i}P_{R_i|\mathcal{E},X_i}\|_1}{2\zeta/3 - \sqrt{2\delta}} + \frac{\|P_{X_i|\mathcal{E}} - P_{X_i}\|_1}{2\zeta/3 - \sqrt{2\delta}}\right] \\
&\le 7\sqrt{2\delta_1}/\zeta.
\end{aligned}$$

By this and(12), $\mathbb{E}_i\|P_{X_iR_i|\mathcal{E}} - P_{X_iR_i|\mathcal{E},1,y^*}\|_1 \le 40\sqrt{2\delta_1}/\zeta$. Define

$$|\phi'\rangle_{X_{\overline{C}}\widetilde{X}_{\overline{C}}Y_{\overline{C}}\widetilde{Y}_{\overline{C}}ABZ_{\overline{C}}|y^*r_i} := \sum_{x_i}\sqrt{P_{X_i|\mathcal{E},r_i}}|\phi\rangle_{X_{\overline{C}}\widetilde{X}_{\overline{C}}Y_{\overline{C}}\widetilde{Y}_{\overline{C}}ABZ_{\overline{C}}|x_iy^*r_i}.$$

It is the same as $|\phi\rangle_{y^*r_i}$ with X_i's distribution unconditioned on $Y_i = y^*$. By(14) and Fact23,

$$\mathbb{E}_iP_{X_iR_i|\mathcal{E}}\left(S^{\zeta}_{\infty}(\phi'_{Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}}|x_iy^*r_i}\|\phi'_{Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}}|y^*r_i}) > \frac{28(2c+\delta_1)+1}{\zeta^4}\right) \le \zeta + \frac{20\sqrt{2\delta_1}}{\zeta}.$$

Fact25 gives projectors on $X_{\overline{C}}\widetilde{X}_{\overline{C}}A$ succeeding with probability $\alpha = 2^{-c'}$, $c' = 60c/\zeta^5$, on $|\phi'\rangle_{X_{\overline{C}}\widetilde{X}_{\overline{C}}Y_{\overline{C}}\widetilde{Y}_{\overline{C}}ABZ_{\overline{C}}|y^*r_i}$, and

$$\begin{aligned}
\mathbb{E}_i\mathbb{E}_{P_{X_iR_i|\mathcal{E}}}\left\|\frac{1}{\alpha}(\Pi_{x_ir_i}\otimes 1)|\phi'\rangle\langle\phi'|_{y^*r_i}(\Pi_{x_ir_i}\otimes 1) - |\phi\rangle\langle\phi|_{x_iy^*r_i}\right\|_1 \\
\le 3\zeta + \frac{40\sqrt{2\delta_1}}{\zeta^2} \le \frac{7\zeta}{2}. \quad (15)
\end{aligned}$$

Complete author Lemma32 proof, unitaries and recombination

Next construct $U_{y_ir_i}$ taking $|\phi\rangle_{y^*r_i,D_i=1,G_i=y_i}$ close to $|\phi\rangle_{y_ir_i,D_i=1,G_i=y_i}$. Arguments like those leading to(14), on Bob's side without the initial message-information step, upper-bound

$$\mathbb{E}_{P_{X_CY_CZ_CDG|\mathcal{E}}}S(\phi_{Y_{\overline{C}}X_{\overline{C}}\widetilde{X}_{\overline{C}}A|x_Cy_Cz_Cdg}\|\rho_{Y_{\overline{C}}X_{\overline{C}}\widetilde{X}_{\overline{C}}A|x_Cy_Cdg}).$$

Raz's lemma gives

$$\begin{aligned}
2\delta_1 &\ge \mathbb{E}_i\mathbb{E}_{P_{D_iG_iR_i|\mathcal{E}}}I(Y_i : X_{\overline{C}}\widetilde{X}_{\overline{C}}A)_{\phi_{d_ig_ir_i}} \\
&\ge \mathbb{E}_i\tfrac{1}{2}(1-\zeta-\sqrt{2\delta})\mathbb{E}_{P_{R_iG_i|\mathcal{E},D_i=1,G_i\neq y^*}}I(Y_i : X_{\overline{C}}\widetilde{X}_{\overline{C}}A)_{\phi_{r_i,D_i=1,g_i}} \\
&= \mathbb{E}_i\tfrac{1}{2}(1-\zeta-\sqrt{2\delta})\mathbb{E}_{P_{R_iG_iY_i|\mathcal{E},D_i=1,G_i\neq y^*}}S(\phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|y_i,D_i=1,g_i}\|\phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|D_i=1,g_i}) \\
&\ge \mathbb{E}_i\tfrac{1}{2}(1-\zeta-\sqrt{2\delta})\sum_{y_i\in\mathcal{Y}}\mathbb{E}_{P_{R_i|\mathcal{E},D_i=1,G_i=y_i}}P_{G_i|\mathcal{E},D_i=1}(y_i) \\
&\quad\cdot\Big[(1-\zeta-\sqrt{2\delta})\|\phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|y_i,r_i,1,y_i} - \phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|r_i,1,y_i}\|_1^2 \\
&\qquad +(\zeta/3-\sqrt{2\delta})\|\phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|y^*,r_i,1,y_i} - \phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|r_i,1,y_i}\|_1^2\Big].
\end{aligned}$$

Use(9) and Pinsker in the last line. By triangle inequality,

$$\begin{aligned}
&\mathbb{E}_i \sum_{y_i\in\mathcal{Y}} \mathbb{E}_{P_{R_i|\mathcal{E},1,y_i}} P_{G_i|\mathcal{E},1}(y_i) \|\phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|y_i r_i,1,y_i} - \phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|y^* r_i,1,y_i}\|_1^2 \\
&\le \mathbb{E}_i \sum_{y_i\in\mathcal{Y}} \mathbb{E}_{P_{R_i|\mathcal{E},1,y_i}} 2P_{G_i|\mathcal{E},1}(y_i) \\
&\quad \cdot [\|\phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|y_i,r_i,1,y_i} - \phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|r_i,1,y_i}\|_1^2 + \|\phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|y^*,r_i,1,y_i} - \phi_{X_{\overline{C}}\widetilde{X}_{\overline{C}}A|r_i,1,y_i}\|_1^2] \\
&\le \frac{4\delta_1}{1-\zeta-\sqrt{2\delta}}\left(\frac{1}{1-\zeta-\sqrt{2\delta}} + \frac{1}{\zeta/3-\sqrt{2\delta}}\right) \le \frac{32\delta_1}{\zeta}.
\end{aligned}$$

The full conditional states are pure, so Fuchs–van de Graaf and Uhlmann give unitaries on $Y_{\overline{C}}\widetilde{Y}_{\overline{C}}BZ_{\overline{C}}$ with

$$\begin{aligned}
&\mathbb{E}_i \sum_{y_i\in\mathcal{Y}} \mathbb{E}_{P_{R_i|\mathcal{E},1,y_i}} P_{G_i|\mathcal{E},1}(y_i) \\
&\quad \cdot \||\phi\rangle\langle\phi|_{y_i r_i,1,y_i} - (1\otimes U_{y_i r_i})|\phi\rangle\langle\phi|_{y^* r_i,1,y_i}(1\otimes U_{y_i r_i}^\dagger)\|_1 \le (32\delta_1/\zeta)^{1/4}. \quad (16)
\end{aligned}$$

It remains to show $\Pi_{x_i r_i}\otimes U_{y_i r_i}$ maps $|\phi'\rangle_{y^* r_i}$ close to $|\phi\rangle_{x_i y_i r_i}$. First show the unitary takes $|\phi\rangle_{x_i y^* r_i}$ close to $|\phi\rangle_{x_i y_i r_i}$. Let O_{X_i} measure X_i and write its outcome in another register. Then

$$\begin{aligned}
O_{X_i}(|\phi\rangle\langle\phi|_{y_i r_i,1,y_i}) &= \sum_{x_i} P_{X_i|\mathcal{E},y_i r_i,D_i=1,G_i=y_i}(x_i)|x_i\rangle\langle x_i| \otimes |\phi\rangle\langle\phi|_{x_i y_i r_i,1,y_i} \\
&= \sum_{x_i} P_{X_i|\mathcal{E},y_i r_i,D_i=1,G_i=y_i}(x_i)|x_i\rangle\langle x_i| \otimes |\phi\rangle\langle\phi|_{x_i y_i r_i}, \\
O_{X_i}(|\phi\rangle\langle\phi|_{y^* r_i,1,y_i}) &= \sum_{x_i} P_{X_i|\mathcal{E},y^* r_i,D_i=1,G_i=y_i}(x_i)|x_i\rangle\langle x_i| \otimes |\phi\rangle\langle\phi|_{x_i y^* r_i}.
\end{aligned}$$

The extra D_i, G_i conditions do not change the two fully conditioned states. Fact10 gives

$$\mathbb{E}_i \|P_{X_i G_i R_i|\mathcal{E},1} - P_{G_i R_i|\mathcal{E},1} P_{X_i|1,G_i}\|_1 \le 2\sqrt{2\delta_1}.$$

Hence for any value $Y_i = y_i$,

$$\begin{aligned}
&\mathbb{E}_i \|P_{X_i G_i R_i|\mathcal{E},1} - P_{G_i R_i|\mathcal{E},1} P_{X_i|\mathcal{E},y_i,1,G_i R_i}\|_1 \\
&\le \mathbb{E}_i [\|P_{X_i G_i R_i|\mathcal{E},1} - P_{G_i R_i|\mathcal{E},1} P_{X_i|y_i,1,G_i}\|_1 \\
&\qquad + \|P_{G_i R_i|\mathcal{E},1}(P_{X_i|y_i,1,G_i} - P_{X_i|\mathcal{E},y_i,1,G_i R_i})\|_1] \\
&\le \mathbb{E}_i \left[\|P_{X_i G_i R_i|\mathcal{E},1} - P_{G_i R_i|\mathcal{E},1} P_{X_i|1,G_i}\|_1 \right. \\
&\qquad \left. + \frac{2}{\zeta/3-\sqrt{2\delta}} \|P_{X_i G_i R_i|\mathcal{E},1} - P_{G_i R_i|\mathcal{E},1} P_{X_i|1,G_i}\|_1\right] \\
&\le 8\sqrt{2\delta_1}/\zeta.
\end{aligned}$$

Here $P_{Y_i|1,g_i}(y_i) \ge \zeta/3 - \sqrt{2\delta}$ for every $G_i = g_i$. Also

$$\mathbb{E}_i \|P_{X_i G_i R_i|\mathcal{E},1} - P_{X_i Y_i R_i|\mathcal{E},1}\|_1 \le 2P_{Y_i G_i|\mathcal{E},1}(Y_i \ne G_i) \le \zeta/3 + \sqrt{2\delta}.$$

Since $P_{X_iY_iR_i|\mathcal{E},1} = P_{X_iY_iR_i|\mathcal{E}}$,

$$\mathbb{E}_i \| P_{X_iY_iR_i|\mathcal{E}} - P_{G_iR_i|\mathcal{E},1} P_{X_i|\mathcal{E},y_i,1,G_iR_i} \|_1 \leq 8\sqrt{2\delta_1}/\zeta + \zeta/3 + \sqrt{2\delta}.$$

Using this with(16),

$$\begin{aligned}
&\mathbb{E}_i \mathbb{E}_{P_{X_iY_iR_i|\mathcal{E}}} \| |\phi\rangle\langle\phi|_{x_iy_ir_i} - (1 \otimes U_{y_ir_i})|\phi\rangle\langle\phi|_{x_iy^*r_i}(1 \otimes U_{y_ir_i}^\dagger) \|_1 \\
&\leq \mathbb{E}_i \Big[\| P_{X_iY_iR_i|\mathcal{E}} - P_{G_iR_i|\mathcal{E},1} P_{X_i|\mathcal{E},y_i,1,G_iR_i} \|_1 \\
&\quad + \| P_{X_iY_iR_i|\mathcal{E}} - P_{G_iR_i|\mathcal{E},1} P_{X_i|\mathcal{E},y^*,1,G_iR_i} \|_1 \\
&\quad + \mathbb{E}_{P_{G_iR_i|\mathcal{E},1}} \Big\| \mathbb{E}_{P_{X_i|\mathcal{E},y_ir_i,1,y_i}} |x_i\rangle\langle x_i| \otimes |\phi\rangle\langle\phi|_{x_iy_ir_i} \\
&\qquad - \mathbb{E}_{P_{X_i|\mathcal{E},y^*r_i,1,y_i}} (1 \otimes U_{y_ir_i})|\phi\rangle\langle\phi|_{x_iy^*r_i}(1 \otimes U_{y_ir_i}^\dagger) \Big\|_1 \Big] \\
&= 16\sqrt{2\delta_1}/\zeta + 2\zeta/3 + 2\sqrt{2\delta} + (32\delta_1/\zeta)^{1/4} < 7\zeta/10. \qquad (17)
\end{aligned}$$

The last term is bounded by Fact15 applied to(16) with O_{X_i}. Conditioning on $G_i \neq y^*$ has also been removed: if $G_i = y^*$, both states are $|\phi\rangle_{x_iy^*r_i}$. From(15),(17), triangle inequality gives

$$\begin{aligned}
&\mathbb{E}_i \mathbb{E}_{P_{X_iY_iR_i|\mathcal{E}}} \left\| \frac{1}{\alpha} (\Pi_{x_ir_i} \otimes U_{y_ir_i})|\phi'\rangle\langle\phi'|_{y^*r_i} (\Pi_{x_ir_i} \otimes U_{y_ir_i}^\dagger) - |\phi\rangle\langle\phi|_{x_iy_ir_i} \right\|_1 \\
&\leq \mathbb{E}_i \mathbb{E}_{P_{X_iY_iR_i|\mathcal{E}}} \Big[\Big\| \frac{1}{\alpha} (\Pi_{x_ir_i} \otimes U_{y_ir_i})|\phi'\rangle\langle\phi'|_{y^*r_i} (\Pi_{x_ir_i} \otimes U_{y_ir_i}^\dagger) \\
&\qquad - (1 \otimes U_{y_ir_i})|\phi\rangle\langle\phi|_{x_iy^*r_i}(1 \otimes U_{y_ir_i}^\dagger) \Big\|_1 \\
&\quad + \|(1 \otimes U_{y_ir_i})|\phi\rangle\langle\phi|_{x_iy^*r_i}(1 \otimes U_{y_ir_i}^\dagger) - |\phi\rangle\langle\phi|_{x_iy_ir_i}\|_1 \Big] \\
&= \mathbb{E}_i \mathbb{E}_{P_{X_iY_iR_i|\mathcal{E}}} \Big[\Big\| \frac{1}{\alpha} (\Pi_{x_ir_i} \otimes 1)|\phi'\rangle\langle\phi'|_{y^*r_i} (\Pi_{x_ir_i} \otimes 1) - |\phi\rangle\langle\phi|_{x_iy^*r_i} \Big\|_1 \\
&\quad + \|(1 \otimes U_{y_ir_i})|\phi\rangle\langle\phi|_{x_iy^*r_i}(1 \otimes U_{y_ir_i}^\dagger) - |\phi\rangle\langle\phi|_{x_iy_ir_i}\|_1 \Big] \\
&\leq 7\zeta/2 + 7\zeta/10 = 21\zeta/5. \qquad (18)
\end{aligned}$$

Markov's inequality on(11),(13),(18) gives an index $i \in \overline{C}$ satisfying all three conditions of Lemma32. This completes its proof, and hence the author's direct-product proof. □

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